

# IBM Granite New Model Variants: Feb 1 – May 25, 2026

## APEX AI Race Engineer — IBM SkillsBuild AI Builders Challenge Reference

**Scope:** Web-grounded enumeration of every IBM Granite model variant released or materially updated between 2026-02-01 and 2026-05-25, with deployment metadata for the APEX 12-tool stack audit. Models are listed in release-date order. Models already present in the APEX inventory are flagged; net-new candidates are called out explicitly at the bottom.

### Summary Table

| #  | Model                      | Release Date | Params | HF URL                                                 | OpenRouter | <a href="#">watsonx.ai</a> | Code Engine     | Vercel Gateway | License    |
|----|----------------------------|--------------|--------|--------------------------------------------------------|------------|----------------------------|-----------------|----------------|------------|
| 1  | Granite 4.0 1B Speech      | 2026-03-06   | ~1B    | ibm-granite/granite-4.0-1b-speech                      | No         | No                         | Yes (container) | No             | Apache 2.0 |
| 2  | FlowState r1.1             | ~2026-03-31  | 18.5M  | ibm-granite/granite-timeseries-flowstate-r1 (rev=r1.1) | No         | No                         | Yes (container) | No             | Apache 2.0 |
| 3  | Granite 4.1 3B             | 2026-04-29   | 3B     | ibm-granite/granite-4.1-3b                             | No         | Yes                        | Yes (container) | No             | Apache 2.0 |
| 4  | Granite 4.1 8B             | 2026-04-29   | 8B     | ibm-granite/granite-4.1-8b                             | Yes        | Yes                        | Yes (container) | No             | Apache 2.0 |
| 5  | Granite 4.1 30B            | 2026-04-29   | 30B    | ibm-granite/granite-4.1-30b                            | No         | Yes                        | Yes (container) | No             | Apache 2.0 |
| 6  | Granite Vision 4.14B       | 2026-04-29   | 4B     | ibm-granite/granite-vision-4.1-4b                      | No         | Yes                        | Yes (container) | No             | Apache 2.0 |
| 7  | Granite Speech 4.1 2B      | 2026-04-29   | ~2B    | ibm-granite/granite-speech-4.1-2b                      | No         | No                         | Yes (container) | No             | Apache 2.0 |
| 8  | Granite Speech 4.1 2B-Plus | 2026-04-29   | ~2B    | ibm-granite/granite-speech-4.1-2b-plus                 | No         | No                         | Yes (container) | No             | Apache 2.0 |
| 9  | Granite Speech 4.1 2B-NAR  | 2026-04-29   | ~2B    | ibm-granite/granite-speech-4.1-2b-nar                  | No         | No                         | Yes (container) | No             | Apache 2.0 |
| 10 | Granite Guardian 4.1 8B    | 2026-04-29   | 8B     | ibm-granite/granite-guardian-4.1-8b                    | No         | No (preview)               | Yes (container) | No             | Apache 2.0 |

| #  | Model                                  | Release Date | Params | HF URL                                             | OpenRouter | <a href="#">watsonx.ai</a> | Code Engine     | Vercel Gateway | License    |
|----|----------------------------------------|--------------|--------|----------------------------------------------------|------------|----------------------------|-----------------|----------------|------------|
| 11 | Granite Embedding 97M Multilingual R2  | 2026-04-29   | 97M    | ibm-granite/granite-embedding-97m-multilingual-r2  | No         | No                         | Yes (container) | No             | Apache 2.0 |
| 12 | Granite Embedding 311M Multilingual R2 | 2026-04-29   | 311M   | ibm-granite/granite-embedding-311m-multilingual-r2 | No         | No                         | Yes (container) | No             | Apache 2.0 |

## Detailed Model Entries

### 1. Granite 4.0 1B Speech

**Release Date:** March 6, 2026<sup>[^1]</sup>

**Version:** 4.0 1B Speech (v1)<sup>[^2]</sup>

**Parameter Count:** ~1 billion<sup>[^2]</sup>

**Hugging Face URL:** <https://huggingface.co/ibm-granite/granite-4.0-1b-speech><sup>[^1]</sup>

**OpenRouter Availability:** No — not listed on OpenRouter at time of research.

**IBM [watsonx.ai](#) Availability:** Not currently listed in the IBM [watsonx.ai](#) multitenant or deploy-on-demand catalog as of the scraped model list. Available via HuggingFace transformers and vLLM self-hosted deployments.<sup>[^3]</sup>

**IBM Cloud Code Engine Compatibility:** Yes — the model ships with native vLLM and Transformers support, meaning it can be containerized and served as a Code Engine application or job.<sup>[^2]</sup>

**Vercel AI Gateway Availability:** No — IBM Granite Speech models do not appear in the Vercel AI Gateway model catalog.<sup>[^4]</sup>

**Intended Use (IBM Release Notes):** Compact, resource-constrained multilingual ASR and bidirectional AST for edge deployments. Achieves top-1 ranking on the OpenASR Leaderboard at release. Adds Japanese ASR and keyword list biasing (names/acronyms) not present in the predecessor Speech 3.3 2B. Half the parameters of Granite Speech 3.3 2B with higher English transcription accuracy via speculative decoding.<sup>[5][2]</sup>

**License + Commercial Terms:** Apache 2.0 — unrestricted research and commercial use.<sup>[^2]</sup>

### 2. FlowState r1.1

**Release Date:** ~March 31, 2026 (tagged as r1.1 revision in the HF repo; the Hugging Face commit dcdc956 updating tags was made March 19, 2026)<sup>[6][7]</sup>

**Version:** r1.1 (incremental update to r1.0 released September 2025)<sup>[^8]</sup>

**Parameter Count:** 18.5M (up from ~9.1M in r1.0; the r1.1 adds a larger MLP layer)<sup>[^8]</sup>

**Hugging Face URL:** <https://huggingface.co/ibm-granite/granite-timeseries-flowstate-r1> — load with revision="r1.1"<sup>[^8]</sup>

**OpenRouter Availability:** No — time-series models are not listed on OpenRouter.

IBM [watsonx.ai](#) Availability: Not in the current publicly documented IBM [watsonx.ai](#) text/chat model catalog. TTM-R2 variants are listed in watsonx for forecasting, but FlowState is not.<sup>[^3]</sup>

IBM Cloud Code Engine Compatibility: Yes — can be containerized via the `granite-tsfm` Python library and served as a Code Engine batch job or app.<sup>[9][10]</sup>

Vercel AI Gateway Availability: No.

Intended Use (IBM Release Notes): r1.1 adds synthetic pre-training data from the CauKer dataset, increases pre-training context from 2,048 to 4,096 tokens, improves the S5 SSM layer with an output gating mechanism, and optimizes hyperparameters. Core use case is **sampling-rate-invariant zero-shot univariate forecasting** across arbitrary temporal resolutions — traffic, energy, sales, IoT telemetry. Remains #2–3 on the GIFT-Eval zero-shot forecasting leaderboard.<sup>[11][8]</sup>

License + Commercial Terms: Apache 2.0.<sup>[^8]</sup>

### 3. Granite 4.1 3B

Release Date: April 29, 2026<sup>[12][13]</sup>

Version: 4.1 3B Instruct (+ 3B Base)<sup>[^14]</sup>

Parameter Count: 3 billion<sup>[^12]</sup>

Hugging Face URL: <https://huggingface.co/ibm-granite/granite-4.1-3b><sup>[^15]</sup>

OpenRouter Availability: Not found as a standalone route on OpenRouter at time of research (only 4.1 8B is listed).<sup>[^16]</sup>

IBM [watsonx.ai](#) Availability: Yes — API model ID `ibm/granite-4-1-3b` listed in the IBM [watsonx.ai](#) foundation models catalog for inferencing (pay-per-token). Available in Dallas and other IBM Cloud regions.<sup>[^17]</sup>

IBM Cloud Code Engine Compatibility: Yes — Apache 2.0 weights support containerization; IBM documents deployment via transformers and vLLM.<sup>[^14]</sup>

Vercel AI Gateway Availability: No IBM Granite models appear in the current Vercel AI Gateway model listings.<sup>[^4]</sup>

Intended Use (IBM Release Notes): Compact edge/resource-constrained enterprise LLM for summarization, classification, extraction, RAG, tool/function-calling, multilingual dialog (12 languages), and FIM code completion. Trained from scratch on ~15 trillion tokens via a 5-phase strategy.<sup>[13][14]</sup>

License + Commercial Terms: Apache 2.0; cryptographically signed; ISO 42001 certified.<sup>[^18]</sup>

### 4. Granite 4.1 8B ☆ (Upgrade target for APEX Granite Instruct 4.1 8B — already in stack)

Release Date: April 29, 2026<sup>[^19]</sup>

Version: 4.1 8B Instruct (+ 8B Base, FP8 quantized variants)<sup>[^20]</sup>

Parameter Count: 8 billion<sup>[^19]</sup>

Hugging Face URL: <https://huggingface.co/ibm-granite/granite-4.1-8b><sup>[^19]</sup>

OpenRouter Availability: Yes — Model ID `ibm-granite/granite-4.1-8b`; \$0.05/1M input, \$0.10/1M output; 131K context window; provider: W&B/CoreWeave.<sup>[21][22]</sup>

IBM [watsonx.ai](#) Availability: Yes — API model ID `ibm/granite-4-1-8b` listed in [watsonx.ai](#) pay-per-token catalog. Available in Dallas, Frankfurt, Sydney, Toronto, and other regional data centers per IBM Cloud

documentation.[^17]

**IBM Cloud Code Engine Compatibility:** Yes — documented native vLLM/transformers support; containerizable.[^14]

**Vercel AI Gateway Availability:** No IBM Granite models appear in the Vercel AI Gateway model listings. [^4]

**Intended Use (IBM Release Notes):** Dense decoder-only 8B instruct LLM, the headline model of the 4.1 family. Benchmarks match or exceed Granite 4.0-H-Small (32B MoE, 9B active) across IFEval, AlpacaEval, MMLU-Pro, BBH, GSM8K, BFCL v3, ArenaHard. Targets RAG, multi-tool agentic workflows, code generation (FIM), multilingual enterprise chatbots, 128K production/512K extended context.[^23][20]

**License + Commercial Terms:** Apache 2.0; cryptographically signed; ISO 42001 certified (first open LM family to hold this designation).[23][18]

## 5. Granite 4.1 30B

**Release Date:** April 29, 2026[^12]

**Version:** 4.1 30B Instruct (+ 30B Base)[^14]

**Parameter Count:** 30 billion[^12]

**Hugging Face URL:** <https://huggingface.co/ibm-granite/granite-4.1-30b>[^12]

**OpenRouter Availability:** Not listed as a separate route on OpenRouter at time of research. [Cloudprice.net](https://cloudprice.net) shows it as "Available" under IBM-Granite but without pricing.[^24]

**IBM [watsonx.ai](https://watsonx.ai) Availability:** Yes — API model ID `ibm/granite-4-1-30b` in the [watsonx.ai](https://watsonx.ai) pay-per-token catalog.[^17]

**IBM Cloud Code Engine Compatibility:** Yes — supports containerization via vLLM and transformers.[25][14]

**Vercel AI Gateway Availability:** No.[^4]

**Intended Use (IBM Release Notes):** Highest-capacity Granite 4.1 model for complex reasoning, specialized tasks, and high-accuracy enterprise automation. Dense decoder-only with 64 layers, 4096 embedding dimension, 32 attention heads, 8 KV heads, MLP hidden 32768. 512K context extension available.[20][23][^14]

**License + Commercial Terms:** Apache 2.0.[^12]

## 6. Granite Vision 4.1 4B ★ (Upgrade of APEX's Granite Vision 4.1 4B — already in stack)

**Release Date:** April 29, 2026[26][27]

**Version:** Vision 4.1 4B[^28]

**Parameter Count:** 4 billion[^26]

**Hugging Face URL:** <https://huggingface.co/ibm-granite/granite-vision-4.1-4b>[^27]

**OpenRouter Availability:** Not listed on OpenRouter at time of research.

**IBM [watsonx.ai](https://watsonx.ai) Availability:** Yes — API model ID `ibm/granite-vision-4-1-4b` listed in the [watsonx.ai](https://watsonx.ai) pay-per-token catalog (chart extraction, table extraction, semantic key-value pair extraction).[^17]

**IBM Cloud Code Engine Compatibility:** Yes — LoRA-based architecture supported on vLLM; containerizable.[^28]

Vercel AI Gateway Availability: No.[^4]

**Intended Use (IBM Release Notes):** Vision-language model for enterprise document data extraction, specifically tuned for table/chart understanding, key-value extraction, and document comprehension. Uses a DeepStack-inspired feature injection distributing visual information across multiple LLM layers for higher fidelity. At 4B parameters, reportedly surpasses Claude-Opus-4.6 on table and graph recognition benchmarks. Paired with a new million-scale chart dataset called ChartNet.[<sup>23</sup>][20][^26]

**License + Commercial Terms:** Apache 2.0.[^28]

## 7. Granite Speech 4.1 2B ★ (Upgrade of APEX's Speech model — predecessor was Speech 3.3 8B)

**Release Date:** April 29, 2026[<sup>29</sup>][30]

**Version:** Speech 4.1 2B (autoregressive)[^31]

**Parameter Count:** ~2 billion[^29]

**Hugging Face URL:** <https://huggingface.co/ibm-granite/granite-speech-4.1-2b>[^30]

**OpenRouter Availability:** Not listed on OpenRouter.

**IBM [watsonx.ai](#) Availability:** Not in the current publicly documented [watsonx.ai](#) catalog for multitenant inference. Available via Azure AI Foundry and self-hosted vLLM.[<sup>32</sup>][3]

**IBM Cloud Code Engine Compatibility:** Yes — transformers and vLLM native support.[^31]

Vercel AI Gateway Availability: No.[^4]

**Intended Use (IBM Release Notes):** Multilingual ASR and AST for enterprise speech pipelines. Reports 5.33% mean WER on the OpenASR Leaderboard. Supports 6 languages for ASR (English, French, German, Spanish, Portuguese, Japanese) and English-to-8-language translation. Trained on 174,000 hours of audio.[<sup>33</sup>][29][^31]

**License + Commercial Terms:** Apache 2.0.[^30]

## 8. Granite Speech 4.1 2B-Plus (NEW variant vs. APEX stack)

**Release Date:** April 29, 2026[^31]

**Version:** Speech 4.1 2B-Plus[^31]

**Parameter Count:** ~2 billion[^31]

**Hugging Face URL:** <https://huggingface.co/ibm-granite/granite-speech-4.1-2b-plus>[^26]

**OpenRouter Availability:** No.

**IBM [watsonx.ai](#) Availability:** Not in current multitenant catalog.[^3]

**IBM Cloud Code Engine Compatibility:** Yes — vLLM/transformers native.[^31]

Vercel AI Gateway Availability: No.[^4]

**Intended Use (IBM Release Notes):** Adds speaker-attributed ASR (speaker diarization), timestamps, and keyword-prompted ASR on top of the base 2B ASR/AST capabilities. Intended for call-center transcription, meeting intelligence, and anywhere speaker identity is required.[<sup>34</sup>][31]

**License + Commercial Terms:** Apache 2.0.[^26]

## 9. Granite Speech 4.1 2B-NAR (NEW variant vs. APEX stack)

Release Date: April 29, 2026<sup>[29]</sup><sup>[31]</sup>

Version: Speech 4.1 2B Non-Autoregressive<sup>[^31]</sup>

Parameter Count: ~2 billion<sup>[^31]</sup>

Hugging Face URL: <https://huggingface.co/ibm-granite/granite-speech-4.1-2b-nar><sup>[^26]</sup>

OpenRouter Availability: No.

IBM [watsonx.ai](#) Availability: Not in current multitenant catalog.<sup>[^3]</sup>

IBM Cloud Code Engine Compatibility: Yes — vLLM/transformers native.<sup>[^31]</sup>

Vercel AI Gateway Availability: No.<sup>[^4]</sup>

**Intended Use (IBM Release Notes):** Non-autoregressive (NLE architecture) ASR variant that trades marginal accuracy for significantly higher throughput and lower latency. Intended for bulk transcription pipelines, real-time ASR with tight SLAs, and high-concurrency speech-to-text workloads.<sup>[34]</sup><sup>[29]</sup>

License + Commercial Terms: Apache 2.0.<sup>[^31]</sup>

## 10. Granite Guardian 4.1 8B ★ (Upgrade of APEX's Granite Guardian 4.1 8B — already in stack)

Release Date: April 29, 2026<sup>[^35]</sup>

Version: Guardian 4.1 8B<sup>[^35]</sup>

Parameter Count: 8 billion (fine-tuned on Granite 4.1 8B base)<sup>[^35]</sup>

Hugging Face URL: <https://huggingface.co/ibm-granite/granite-guardian-4.1-8b><sup>[^36]</sup>

OpenRouter Availability: No tracked provider routes at time of research.<sup>[^35]</sup>

IBM [watsonx.ai](#) Availability: Not in the current public [watsonx.ai](#) inferencing catalog (Guardian 3.x variants are listed; 4.1 not confirmed live as of scrape date). IBM's foundation models page lists it as part of the 4.1 family. Regional availability may vary.<sup>[3]</sup><sup>[17]</sup>

IBM Cloud Code Engine Compatibility: Yes — containerizable via HF transformers and vLLM.<sup>[^36]</sup>

Vercel AI Gateway Availability: No.<sup>[^4]</sup>

**Intended Use (IBM Release Notes):** Safety classifier and risk detector for AI pipelines. Detects harm, social bias, jailbreaking, violence, profanity, sexual content, unethical behavior. Also evaluates RAG hallucination (groundedness, context relevance, answer relevance) and **function-calling hallucination** in agentic workflows. New in 4.1: improved Bring Your Own Criteria (BYOC) for custom judging criteria beyond the default risk taxonomy, and a hybrid thinking mode that can produce detailed reasoning (<think> tags) or fast yes/no. Benchmarks: OOD Safety F1 0.79, RAG hallucination avg BAcc 0.76, BFCL function calling BAcc 0.79.<sup>[36]</sup><sup>[35]</sup>

License + Commercial Terms: Apache 2.0.<sup>[^35]</sup>

## 11. Granite Embedding 97M Multilingual R2 ★ (Upgrade of APEX's Granite Embedding R2 149M+47M — new multilingual tier)

Release Date: April 29, 2026<sup>[37]</sup><sup>[38]</sup>

Version: Embedding Multilingual R2 (97M compact)<sup>[^39]</sup>

Parameter Count: 97 million<sup>[^39]</sup>

Hugging Face URL: <https://huggingface.co/ibm-granite/granite-embedding-97m-multilingual-r2><sup>[^40]</sup>

OpenRouter Availability: No — embedding models are not served on OpenRouter.

IBM [watsonx.ai](#) Availability: Not in the main LLM inferencing catalog. IBM watsonx.data and Milvus integrations are documented separately.<sup>[^3]</sup>

IBM Cloud Code Engine Compatibility: Yes — sentence-transformers compatible; containerizable.<sup>[^38]</sup>

Vercel AI Gateway Availability: No.<sup>[^4]</sup>

Intended Use (IBM Release Notes): Bi-encoder multilingual embedding model for dense retrieval across 200+ languages (52 explicitly trained). Built on ModernBERT architecture with expanded multilingual vocabulary; 32,768-token context (64× expansion over R1's 512-token limit). Produces 384-dimensional embeddings. Achieves #1 on MTEB for multilingual embedding models under 100M parameters. Supports Matryoshka Representation Learning on the 311M variant.<sup>[41]</sup><sup>[42]</sup><sup>[38]</sup><sup>[39]</sup>

License + Commercial Terms: Apache 2.0; enterprise-appropriate training data with governance oversight.<sup>[^39]</sup>

## 12. Granite Embedding 311M Multilingual R2 (NEW — distinct from APEX's English-only R2)

Release Date: April 29, 2026<sup>[43]</sup><sup>[37]</sup>

Version: Embedding Multilingual R2 (311M full-size)<sup>[^39]</sup>

Parameter Count: 311 million<sup>[^37]</sup>

Hugging Face URL: <https://huggingface.co/ibm-granite/granite-embedding-311m-multilingual-r2><sup>[^43]</sup>

OpenRouter Availability: No.

IBM [watsonx.ai](#) Availability: IBM [watsonx.ai](#) documentation references the 311M Multilingual R2 as ranking #1 for multilingual models under 311M parameters. Not in the primary LLM token-inference catalog; available for self-hosted RAG.<sup>[^41]</sup>

IBM Cloud Code Engine Compatibility: Yes — sentence-transformers/TEI compatible.<sup>[^44]</sup>

Vercel AI Gateway Availability: No.<sup>[^4]</sup>

Intended Use (IBM Release Notes): Full-size multilingual bi-encoder for enterprise RAG, semantic search, cross-lingual retrieval, long-document search, and code retrieval. Produces 768-dimensional embeddings; 32K context. Supports Matryoshka Representation Learning for flexible output dimensionality. arXiv paper published 2026-05-13.<sup>[45]</sup><sup>[37]</sup><sup>[^39]</sup>

License + Commercial Terms: Apache 2.0.<sup>[^39]</sup>

## Models Searched But NOT Released in Window (Feb 1 – May 25, 2026)

The following were explicitly searched; no evidence of a new release in the defined window was found:

| Category                                  | Status                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Granite Code (new 4.x code model)         | No new dedicated Code model released in window. Granite 4.1 8B/30B incorporates code capabilities (FIM, code gen, explanation) but there is no separate granite-code-4.x model card released in this period <sup>[46]</sup> . The existing granite-8b-code-instruct remains the standalone code model on watsonx <sup>[3]</sup> . |
| TSPulse r2                                | Not released. TSPulse-r1 was accepted at ICLR 2026 (Mar 2026) <sup>[47]</sup> <sup>[48]</sup> , but no r2 checkpoint has been published.                                                                                                                                                                                          |
| TTM r3                                    | Not released. TTM-R2.1 (released Feb 25, 2025 <sup>[49]</sup> ) remains the current checkpoint. IBM watsonx lists granite-ttm-512-96-r2, granite-ttm-1024-96-r2, granite-ttm-1536-96-r2 as the active TTM variants <sup>[3]</sup> .                                                                                               |
| Granite-Docling r2                        | Not released. Granite-Docling-258M (v1, released September 2025 <sup>[50]</sup> ) remains the only version; no r2 announcement found.                                                                                                                                                                                             |
| Granite Guardian r2 / 4.1 2B              | Granite Guardian 4.1 8B released (entry #10 above); no 2B Guardian variant in the 4.1 family was announced.                                                                                                                                                                                                                       |
| Granite Mellea / Instruct-Validate-Repair | No new release found. IBM Bob (the Instruct-Validate-Repair agent) continues as part of the watsonx Code Assistant ecosystem but no new standalone Granite Mellea model card was published in this window.                                                                                                                        |
| Granite 4.0 Nano new update               | The 4.0 Nano family (H-1B ~1.5B, H-350M ~350M, 1B, 350M) was released October 27, 2025 <sup>[51]</sup> <sup>[52]</sup> — outside the window. No 4.1 Nano variant was released in the Feb–May 2026 window.                                                                                                                         |

### ⚠ APEX Stack Gap Analysis — Best Use of Technology Track

APEX's current 12-tool inventory is: Granite-Docling 258M, Docling library, Granite Vision 4.1 4B, Granite TimeSeries TTM r2.1, Granite FlowState 9.1M, IBM TSPulse 1M, Granite Embedding R2 149M+47M (English), Granite Instruct 4.1 8B, Granite Guardian 4.1 8B, Granite 4.0 Nano 350M, LangGraph + MCP Gateway + ContextForge, IBM Bob.

#### Gap 1 — FlowState r1.1 (18.5M) ← ADD THIS

**Justification:** APEX currently runs FlowState at the r1.0 checkpoint (9.1M params); the r1.1 drop (March 2026) adds output gating, 4096 pre-training context, CauKer synthetic data, and a 2× larger MLP — delivering meaningfully improved zero-shot lap-time interval forecasting with no API or license change required.

#### Gap 2 — Granite Speech 4.1 2B-Plus (speaker-attributed ASR) ← Consider Adding

**Justification:** APEX currently uses a generic speech model (implied by the stack), but Granite Speech 4.1 2B-Plus adds speaker diarization and timestamp attribution — directly useful for in-car multi-crew radio transcription (driver + race engineer + spotter) without any external diarization service.

#### Gap 3 — Granite Embedding 311M Multilingual R2 ← Consider Adding

**Justification:** APEX's current Embedding R2 (149M + 47M) is English-only; for international adaptive racing events (F1, Formula E, touring cars), switching to the 311M Multilingual R2 adds 200+ language retrieval and a 32K token context window — enabling cross-lingual RAG over race regulations and driver debriefs in any language.

## Gap 4 — Granite 4.1 30B ← Optional / Challenge Demo

**Justification:** APEX's primary LLM is Granite 4.1 8B; adding the 30B as a fallback reasoning tier for complex adaptive-driving regulation interpretation or post-session strategy review would signal architectural depth to the "Best Use of Technology" judges without changing the production cost profile.

## Deployment Notes for APEX

### watsonx.ai Regions (Confirmed Models)

The following Granite 4.1 models are confirmed live on IBM watsonx.ai as of May 2026:[<sup>17</sup>]

- `ibm/granite-4-1-3b` — 131K context, pay-per-token
- `ibm/granite-4-1-8b` — 131K context, pay-per-token
- `ibm/granite-4-1-30b` — 131K context, pay-per-token
- `ibm/granite-vision-4-1-4b` — 131K context, pay-per-token

IBM watsonx regional availability varies by data center (Dallas, Frankfurt, Sydney, Toronto, London). Dallas users can follow model card links directly.[<sup>3</sup>]

### OpenRouter (IBM-Granite namespace)

As of May 2026, only two IBM Granite models are routed on OpenRouter:[<sup>16</sup>]

- `ibm-granite/granite-4.1-8b` — \$0.05/\$0.10 per 1M tokens, 131K context, W&B provider[<sup>21</sup>]
- `ibm-granite/granite-4.0-micro` (H-Micro hybrid) — \$0.017/\$0.112 per 1M tokens[<sup>16</sup>]

### IBM Cloud Code Engine

All Granite 4.1 models (language, vision, speech, embedding, guardian) support containerized deployment on IBM Cloud Code Engine via Docker images using the `vllm` or `sentence-transformers` base images. Code Engine's serverless GPU fleet support (announced April 2026) enables on-demand inference scaling to zero — well-suited for the APEX competition demo environment.[<sup>53</sup>][<sup>10</sup>][<sup>9</sup>]

### Vercel AI Gateway

As of May 25, 2026, no IBM Granite models appear in the Vercel AI Gateway catalog. The IBM watsonx-ai provider package provides a Vercel AI SDK adapter (`@ibm/watsonx-ai-provider`), which allows any watsonx-hosted Granite model to be used through the Vercel AI SDK's OpenAI-compatible interface — but this is not the same as the Vercel AI Gateway (which routes through Vercel's managed endpoints). APEX can integrate Granite via the watsonx adapter locally in the Vercel AI SDK.[<sup>54</sup>][<sup>4</sup>]

### License Summary

All models in this report are released under Apache 2.0 — free for research and commercial use, no royalty, no copyleft. IBM additionally provides:[<sup>55</sup>][<sup>13</sup>][<sup>18</sup>]

- Cryptographic signing (from April 29, 2026 for all 4.1 models)[<sup>18</sup>]
- ISO 42001 certification for responsible AI management[<sup>23</sup>]

- IBM indemnification only when models are accessed through [watsonx.ai](https://watsonx.ai) (not through Hugging Face directly)[^3]

## References

1. [ibm-granite/granite-4.0-1b-speech - Hugging Face](#) - Release Date: March 6, 2026. License: Apache 2.0. Supported Languages ... Model tree for ibm-granite...
2. [Granite 4.0 1B Speech: Compact, Multilingual, and Built for the Edge](#) - Like all Granite models, Granite 4.0 1B Speech is released under an Apache 2.0 license with native s...
3. [Supported foundation models in watsonx.ai - IBM Cloud Pak for Data](#) - The following table lists the supported IBM foundation models that IBM provides for inferencing. You...
4. [Browse AI Gateway Models - Vercel](#) - Compare capabilities, pricing, and performance across leading AI providers.
5. [IBM Releases Granite 4.0 1B Speech Model for Multilingual Speech...](#) - IBM just released Granite 4.0 1B Speech, a compact and efficient speech-language model, designed for...
6. [Updated tags · ibm-granite/granite-timeseries-flowstate-r1 at dc956](#) - We're on a journey to advance and democratize artificial intelligence through open source and open s...
7. [Hosted Time Series Foundation Models - TSFM.ai](#) - FlowState r1.1 is IBM's March 31, 2026 refresh of the sampling-rate-invariant FlowState family. Comp...
8. [ibm-research/flowstate - Hugging Face](#) - Total number of parameters: 18.5M. Important: To use FlowState-r1.1 include the revision=r1.1 argue...
9. [Getting started with IBM Cloud Code Engine](#) - Find documentation, API & SDK references, tutorials, FAQs, and more resources for IBM Cloud products...
10. [IBM Cloud Code Engine](#) - IBM Cloud Code Engine is a fully managed, serverless platform. Bring your container images, batch jo...
11. [A more fluid way to model time-series data - IBM Research](#) - The latest addition to IBM's Granite family of time-series models, FlowState is now available in ope...
12. [IBM has released the 'Granite 4.1' series of AI models, open models that excel at following prompts and calling tools.](#) - The news blog specialized in Japanese culture, odd news, gadgets and all other funny stuffs. Updated...
13. [granite4.1 - Ollama](#) - Granite 4.1 language models are a family of state-of-the-art open foundation models featuring dense ...
14. [Granite 4.1 - IBM](#)
15. [ibm-granite/granite-4.1-3b - Hugging Face](#) - Granite-4.1-3B baseline is built on a decoder-only dense transformer architecture. Core components o...
16. [ibm-granite API and Models - OpenRouter](#) - Access 2 ibm-granite models through the OpenRouter unified API including Granite 4.1 8B and Granite ...
17. [Foundation Models - IBM watsonx.ai](#) - IBM Granite family of open, performant and trusted models deliver exceptional performance at a compe...
18. [Granite - IBM](#) - As of 29 April 2026, released Granite language, vision, speech, embedding and guardian models are be...
19. [ibm-granite/granite-4.1-8b · Hugging Face : r/LocalLLaMA - Reddit](#) - Granite 4.1 model release details. Granite 4.1 benchmark performance ... Also IBM: Release Date: Apr...

20. [IBM Releases Granite 4.1: Dense 8B Matches Prior 32B MoE Flagship](#)
21. [Granite 4.1 8B - API Pricing & Benchmarks - OpenRouter](#) - Granite 4.1 8B is a dense, decoder-only 8-billion-parameter language model from IBM, part of the Gra...
22. [IBM: Granite 4.1 8B – Uptime and Availability | OpenRouter](#) - ... 2026. Model weights · Overview · Providers · Performance · Pricing · Benchmarks · Apps · Activit...
23. [IBM Granite 4.1: Open LLMs With 512K Context - Creative AI News](#) - IBM released Granite 4.1 on April 29, 2026 with dense decoder-only 3B, 8B, and 30B Apache 2.0 LLMs t...
24. [Granite 4.1 3B pricing & specs — IBM | CloudPrice](#) - Granite 4.1 8B, Apr 29, 2026, 131K, \$0.050, \$0.100, Available. Granite 4 Micro H, Oct 19, 2025, 131K...
25. [Chat Completions API - AI Model Catalog | Microsoft Foundry Models](#) - Chat Completions API. Send Request. You can use cURL or any REST Client to send a request to the Azu...
26. [IBM has released the 'Granite 4.1' series of AI models ... - Gigazine](#) - The news blog specialized in Japanese culture, odd news, gadgets and all other funny stuffs. Updated...
27. [ibm-granite/granite-vision-4.1-4b - Hugging Face](#) - Granite-Vision-4.1-4B · Developer: IBM Research · Release Date: April 29th, 2026 · License: Apache 2...
28. [Granite Vision Models - GitHub](#) - Latest Release: Granite-Vision-4.1-4B ... Granite-Vision-4.1-4B is a vision-language model tailored ...
29. [IBM Releases Granite Speech 4.1 2B and 2B NAR Models - LinkedIn](#) - IBM just released TWO new open speech recognition models: Granite Speech 4.1 2B Granite Speech 4.1 2...
30. [ibm-granite/granite-speech-4.1-2b - Hugging Face](#) - † We report a Cap-F1 of 99.5 on German, where noun capitalization is required. Release Date: April 2...
31. [Granite Speech - IBM](#) - The Granite Speech 4.1 model family provides compact and efficient speech-language models for multil...
32. [granite-speech-4.1-2b - AI Model Catalog | Microsoft Foundry Models](#) - granite-speech-4.1-2b. Version: 1. Hugging Face • Last updated April 2026. Overview Benchmarks Benchmar...
33. [IBM Releases Two Granite Speech 4.1 2B Models - MarkTechPost](#) - On the Open ASR Leaderboard (as of April 2026), Granite Speech 4.1 2B scores a mean WER of 5.33. Dri...
34. [What Is IBM Granite Speech 4.1? Three ASR Models and When to ...](#) - IBM Granite Speech 4.1 offers three ASR models: a base model, a Plus model with diarization, and a n...
35. [Granite Guardian 4.1 8B – 8K context, open source | LLM Reference](#) - IBM Granite Guardian 4.1 8B is a safety and risk-detection model fine-tuned from Granite 4.1 8B. Fea...
36. [ibm-granite/granite-guardian-4.1-8b - Hugging Face](#) - Granite Guardian 4.1 8B introduces improved Bring Your Own Criteria (BYOC) support, enabling users t...
37. [Granite Embedding 311M Multilingual R2 - LLM Reference](#) - IBM Granite Embedding 311M Multilingual R2 is a text embedding model supporting 200+ languages (with...
38. [ibm-granite/granite-embedding-97m-multilingual-r2 · Hugging Face](#) - IBM just dropped the R2 generation of their Granite multilingual embedding models for retrieval, and...
39. [Granite Embedding Multilingual R2 Models - Cool Papers](#) - We introduce the multilingual Granite Embedding R2 models, a family of encoder-based embedding model...
40. [ibm-granite/granite-embedding-97m-multilingual-r2 - Hugging Face](#) - The model uses a bi-encoder architecture to generate high-quality embeddings from text inputs such a...
41. [Granite Embedding - IBM](#) - Model Collection. View the full Granite Embedding collection on Hugging Face ; Run locally with Olla...

42. [32K Context Hits IBM's Open Multilingual Embedding R2 Models](#) - IBM released Granite Embedding Multilingual R2, upgrading its Apache 2.0 encoder models with a 32,76...
43. [ibm-granite/granite-embedding-311m-multilingual-r2 - Hugging Face](#) - Release Date: April 29, 2026; License: Apache 2.0. Supported Languages. The underlying encoder was p...
44. [Open Apache 2.0 Multilingual Embeddings with 32K Context](#) - Try The Models. Both models are available now on Hugging Face under the IBM Granite Embedding collec...
45. [Granite Embedding Multilingual R2 Models - arXiv](#) - The release consists of two bi-encoder models based on the ModernBERT architecture with an expanded ...
46. [Granite Code](#)
47. [ibm-granite/granite-timeseries-tspulse-r1 - Hugging Face](#) - Update (Mar 2026): TSPulse has been accepted at ICLR 2026. ... Despite being 10-100× smaller than co...
48. [ICLR Poster TSPulse: Tiny Pre-Trained Models with Disentangled ...](#) - TSPulse delivers state-of-the-art zero-shot performance, efficient fine-tuning, and supports GPU-fre...
49. [IBM's next generation Granite models are now available - SD Times](#) - Granite-Timeseries-TTM-R2.1 extends the model's forecasting capabilities to now offer daily and week...
50. [ibm/granite-docling - Ollama](#) - Granite Docling is a multimodal Image-Text-to-Text model engineered for efficient document conversio...
51. [Granite 4.0 Nano: Just how small can you go? - Hugging Face](#) - Today we are excited to share Granite 4.0 Nano, our smallest models yet, released as part of IBM's G...
52. [IBM Unveils Granite 4.0 Nano: Ultra-Compact AI Models for Edge ...](#)
53. [Notas del release para Code Engine](#) - Encuentre documentación, referencias de API y SDK, guías de aprendizaje, preguntas frecuentes y más ...
54. [IBM watsonx.ai Provider for Vercel AI SDK - GitHub](#) - AI SDK is a framework for building AI-powered applications quickly. It can be used to create custom ...
55. [granite-4 - LM Studio](#) - Granite 4.0 language models are lightweight, state-of-the-art open models that natively support mult...